RESEARCH COMPANION

Utilitarian Ethics in AI Alignment

Annotated bibliography and evidence-quality audit

> 63 cited items grouped by argument, annotated for contribution, limits, and research value.

Companion to: “Utilitarian Ethics in Artificial Intelligence: From Philosophical Origins to Benchmarking and Alignment”
Prepared: October 2026

Reader’s guide: the source base is useful but uneven

The report’s argument is supported by a credible core of philosophy reference works, benchmark papers, and formal alignment research, but the 63-item list mixes primary studies with explainers, forum posts, profiles, duplicates, and citations whose role is unclear.

Use first

Normative frame: refs. 3, 5-7, 21, 23.

Benchmarks: refs. 24/25, 30, 32, 34-36.

Alignment mechanics: refs. 39-44, 47, 50, 54, 57, 63.

Use cautiously

Preprints and OpenReview submissions can be technically valuable, but publication status and assumptions matter. Treat references 16, 35, 40, 42, 43, 45, 50, 57, and 62 as provisional until checked.

Replace or remove

Video, Reddit, open threads, profiles, broad resource lists, and tertiary explainers should not carry central claims: refs. 1, 17, 28, 37, 38, 46, 52, 55, 56.

Deduplicate

Refs. 11/21, 16/41, and 24/25 point to the same underlying works or versions. Merge them in a revised bibliography.

How to read the tiers

Tier A peer-reviewed, archival, or foundational scholarly source. Tier B primary technical work, preprint, working paper, or repository manuscript. Tier C authoritative secondary, implementation, or educational source. Tier D informal, tertiary, commercial, biographical, or bibliographically unclear source.

Tiers assess suitability for supporting the report, not the truth of every claim in a source. Annotations are keyed to the numbered “Works cited” list in the supplied report.

Foundations of utilitarian ethics

01

What is Utilitarianism?

Tier D YouTube video

A short orientation to the basic maximizing-consequences idea. It may help a first-time reader, but it is not an appropriate authority for historical or technical claims; use the Stanford Encyclopedia entries below instead.

https://www.youtube.com/watch?v=7rEKfJMe7eM

02

Jeremy Bentham

Tier C Utilitarianism.net profile

A specialist public-education profile used for Bentham's principle of utility, impartiality, and concern for suffering. Useful for accessible context; historical claims should be cross-checked against a scholarly reference work or Bentham's primary text.

https://utilitarianism.net/utilitarian-thinker/jeremy-bentham/

03

The History of Utilitarianism

Tier A Stanford Encyclopedia of Philosophy

The strongest overview in the report's historical section. It situates Bentham, Mill, and later developments in a peer-edited philosophical reference and should anchor claims about the evolution from classical to modern utilitarianism.

https://plato.stanford.edu/entries/utilitarianism-history/

04

Utilitarianism

Tier C CUNY Manifold open textbook

An introductory ethics chapter that clarifies act and rule utilitarianism for non-specialists. It is pedagogically useful but should not carry contested claims about justice, aggregation, or AI implementation.

https://cuny.manifoldapp.org/read/introduction-to-philosophy-ethics/section/f3673367-ced9-42e8-81cc-4590e2f04774

Foundations of utilitarian ethics (continued)

05

Mill’s Moral and Political Philosophy

Tier A Stanford Encyclopedia of Philosophy

An authoritative synthesis of Mill’s qualitative hedonism, liberty, justice, and rule-like reasoning. It directly supports the report’s claim that utility can justify stable protections rather than case-by-case maximization alone.

https://plato.stanford.edu/entries/mill-moral-political/

06

Jeremy Bentham

Tier A Stanford Encyclopedia of Philosophy

A scholarly reference for Bentham’s moral psychology, legal reform, and utility principle. Prefer this source over the profile and video when discussing felicific calculation, equality of consideration, or Bentham’s historical influence.

https://plato.stanford.edu/entries/bentham/

07

John Stuart Mill

Tier A Stanford Encyclopedia of Philosophy

A broad scholarly biography and philosophical overview. It complements reference 5 but is partly redundant; cite the more specific Mill ethics entry when the claim concerns higher pleasures or justice.

https://plato.stanford.edu/entries/mill/

08

Utilitarianism of Mill and Bentham: A Comparative Analysis

Tier D Francis Press PDF

A direct comparison of Bentham and Mill that likely informed the report’s side-by-side table. The source list does not supply authors or venue details, so treat it as provisional and verify publication quality before relying on it.

https://francis-press.com/uploads/papers/KsRrRV5TXue1Ia4acHCmbrHXawX5xjeKJxEuW8T9.pdf

Foundations of utilitarian ethics (continued)

09

Utilitarian Thinkers

Tier C Utilitarianism.net overview

A navigation page surveying major figures and variants. It is useful for finding names and themes, but its broad scope makes it a starting point rather than evidence for any specific doctrinal claim.

https://utilitarianism.net/utilitarian-thinkers/

10

The Nature of Preferences

Tier C RLHF and Post-Training Book chapter

A technical-practitioner discussion of how preferences are elicited and modeled in post-training. It supports the bridge from preference utilitarianism to pairwise comparisons, while remaining a secondary synthesis rather than a single empirical study.

https://rlhfbook.com/c/10-preferences

11

Artificial Intelligence, Values, and Alignment

Tier B ResearchGate copy

A copy of Iason Gabriel's value-alignment paper. The work is central to the report's taxonomy of instructions, intentions, preferences, interests, and values; reference 21 is the cleaner archival record and should be cited instead.

https://www.researchgate.net/publication/345238301_Artificial_Intelligence_Values_and_Alignment

12

Artificial Intelligence and the Economics of Decision-Making

Tier B EconStor discussion paper

A working paper connecting AI with economic decision theory. It is relevant to expected utility and rational-agent modeling, but the report should identify the precise proposition it supports rather than using it as general background.

https://www.econstor.eu/bitstream/10419/272627/1/dp16000.pdf

Foundations of utilitarian ethics (continued)

13

Harsanyi’s Social Aggregation Theorem and what it means for CEV

Tier D AI Alignment Forum post

An informal interpretation of Harsanyi’s aggregation result for coherent extrapolated volition. It is useful for alignment-specific intuition, but the theorem’s assumptions and proof should be sourced to Harsanyi or a scholarly social-choice treatment.

https://www.alignmentforum.org/posts/z8afQRsH9wWsB4iMD/harsanyi-s-social-aggregation-theorem-and-what-it-means-for

14

Towards a scale-free theory of intelligent agency

Tier D LessWrong post

A community essay on agency and utility at different scales. It can generate hypotheses, but it should not be treated as validation of the report’s claim that utility maximization is structurally inevitable for AI.

https://www.lesswrong.com/posts/5tYTKX4pNpiG4vzYg/towards-a-scale-free-theory-of-intelligent-agency

15

Challenges of Aligning Artificial Intelligence with Human Values

Tier D ResearchGate record

A broad alignment source whose bibliographic details are incomplete in the report. Verify authorship, venue, and version before citation; use it only after mapping a specific report claim to a passage in the underlying work.

https://www.researchgate.net/publication/346964247_Challenges_of_Aligning_Artificial_Intelligence_with_Human_Values

Value alignment and social aggregation

16

Distortion of AI Alignment Revisited: RLHF Is a Decent Utilitarian

Tier B OpenReview

A primary technical submission analyzing preference aggregation and distortion in reinforcement learning from human feedback (RLHF). It is highly relevant to the report's argument that a learned reward model acts like a social-welfare aggregator; check the final publication status and theorem conditions.

https://openreview.net/forum?id=BciUs38t6W

17

Open Thread 436

Tier D Astral Codex Ten

A general discussion thread with no stable scholarly claim identified by the report. It should be removed unless a specific comment is quoted and independently supported; as listed, it adds little evidentiary value.

https://www.astralcodexten.com/p/open-thread-436

18

Reward hacking doesn't show reward is the optimization target

Tier D LessWrong post

An informal conceptual critique relevant to how reward optimization should be interpreted. It can sharpen terminology, but it should be paired with primary reinforcement-learning literature before supporting claims about model objectives.

https://www.lesswrong.com/posts/dqSwccGTWyBgxrR58/reward-hacking-doesnt-show-reward-is-optimization-target

19

Complete Class: Consequentialist Foundations

Tier D AI Alignment Forum post

A community-level treatment of consequentialist representation. It is useful for conceptual framing, but the report should not infer that every capable AI is literally a utilitarian agent from representation theorems alone.

https://www.alignmentforum.org/posts/sZuw6SGfmZHvcAAEP/complete-class-consequentialist-foundations

Value alignment and social aggregation (continued)

20

What counts as defection?

Tier D AI Alignment Forum post

An informal discussion of cooperation and defection. Its relevance to the report is peripheral; retain only if the text explicitly analyzes multi-agent utility or cooperative behavior.

https://www.alignmentforum.org/posts/8LEPDY36jBYpijrSw/what-counts-as-defection

21

Artificial Intelligence, Values and Alignment

Tier B arXiv:2001.09768

Gabriel’s primary paper is the key normative source for the report. It distinguishes possible targets of alignment and shows why ‘human values’ is not a single operational objective; it should anchor the report’s taxonomy and its critique of revealed-preference alignment.

https://arxiv.org/abs/2001.09768

22

Thoughts on Iason Gabriel’s Artificial Intelligence, Values, and Alignment

Tier D AI Alignment Forum post

A commentary on reference 21. It may expose objections or extensions, but it is secondary and should not substitute for Gabriel’s own definitions.

https://www.alignmentforum.org/posts/Z2rkdEAJ9MvYPBeYW/thoughts-on-iason-gabriel-s-artificial-intelligence-values

23

Beyond Preferences in AI Alignment

Tier A University of Cambridge repository paper

A scholarly source challenging preference-satisfaction as a complete alignment target. It directly supports the report’s move toward interests, autonomy, and objective well-being, and helps prevent RLHF from being treated as synonymous with moral alignment.

https://www.repository.cam.ac.uk/bitstreams/7969c965-8d16-4080-88d0-1d9250bf3a76/download

Value alignment and social aggregation (continued)

24

Aligning AI With Shared Human Values

Tier B arXiv HTML

The ETHICS benchmark paper in a readable HTML version. It provides the dataset design, normative subsets, and empirical results underlying the report’s discussion; reference 25 is the same work’s abstract record, so only one should remain.

https://arxiv.org/html/2008.02275v6

25

Aligning AI With Shared Human Values

Tier B arXiv:2008.02275

The archival abstract page for the same ETHICS paper as reference 24. Use this canonical record in the bibliography and avoid counting both entries as independent evidence.

https://arxiv.org/abs/2008.02275

Moral benchmarks and behavioral evaluation

26

EleutherAI/hendrycks_ethics dataset script

Tier C Hugging Face dataset repository

Implementation-level documentation for loading ETHICS. It is valuable for reproducibility and subset inspection, but dataset statistics and validity claims should come from the primary paper and later audits.

https://huggingface.co/datasets/EleutherAI/hendrycks_ethics/blob/main/hendrycks_ethics.py

27

Moral AI with Aggregating Models Based on Normative Ethical Theory

Tier B CEUR Workshop Proceedings paper

A workshop paper on combining models associated with normative theories. It is relevant to the report's recommendation for pluralistic or hybrid evaluation; its workshop status warrants treating conclusions as exploratory.

https://ceur-ws.org/Vol-3547/paper3.pdf

28

ETHICS Benchmark for AI Ethics

Tier D Emergent Mind explainer

A tertiary overview of ETHICS. It is useful for quick orientation but should not support quantitative claims, model rankings, or judgments about benchmark validity.

https://www.emergentmind.com/topics/ethics-benchmark

29

arXiv:2402.11777v1

Tier B arXiv preprint PDF

A primary preprint cited without its title in the report. Because the source list does not say which claim it supports, verify the paper's title, authors, and role before retaining it in a revised report.

https://arxiv.org/pdf/2402.11777

Moral benchmarks and behavioral evaluation (continued)

30

Is ETHICS about ethics? Evaluating the ETHICS benchmark

Tier B ar5iv rendering of arXiv:2410.13009

A critical audit of whether ETHICS measures ethical competence rather than pattern recognition or folk judgment. It is essential counterevidence to the report's positive benchmark narrative and supports the distinction between static classification and moral action.

https://ar5iv.labs.arxiv.org/html/2410.13009

31

The MACHIAVELLI Benchmark

Tier C Project page

The authors' project page for the interactive benchmark, with task description and supporting resources. Use it for navigation and demonstrations, but cite the primary paper for design, annotations, and empirical trade-offs.

https://aypan17.github.io/machiavelli/

32

Do the Rewards Justify the Means? Measuring Trade-Offs Between Rewards and Ethical Behavior in the MACHIAVELLI Benchmark

Tier B arXiv:2304.03279

The primary MACHIAVELLI paper. It directly supports the report's claim that goal-directed agents can trade ethical behavior for reward in rich text environments; the experimental setup should not be generalized automatically to all deployed agents.

https://arxiv.org/abs/2304.03279

33

Porting MACHIAVELLI to Inspect

Tier D Koby Lewis technical blog

An implementation note on adapting the benchmark to an evaluation framework. It is useful for replication work but not for claims about the benchmark's validity or the inevitability of deception.

https://kobylewis.net/ai/Porting-MACHIAVELLI-to-Inspect

Moral benchmarks and behavioral evaluation (continued)

34

AI deception: A survey of examples, risks, and potential solutions

Tier A PubMed Central review

A broad scholarly review of deceptive AI behavior and mitigation. It supports the report's risk discussion, while also supplying nuance: deception has multiple mechanisms and should not be reduced to a simple by-product of utilitarianism.

https://pmc.ncbi.nlm.nih.gov/articles/PMC11117051/

35

LLM Ethics Benchmark: A Three-Dimensional Assessment System

Tier B arXiv:2505.00853v1

A recent preprint proposing multidimensional ethics evaluation. It supports the report's call to assess reasoning, consistency, and value pluralism beyond a single utility score; results remain provisional until peer review and independent replication.

https://arxiv.org/html/2505.00853v1

36

Exploring Multilingual Concepts of Human Values in Large Language Models

Tier A Findings of EMNLP 2024

A peer-reviewed empirical study on value concepts across languages. It strengthens the report's concern that moral benchmarks built in one language or culture may not capture globally shared or culturally variable values.

https://aclanthology.org/2024.findings-emnlp.96.pdf

Preference optimization and RLHF

37

What Is RLAIF in AI Alignment & How Does It Work?

Tier D Mercor explainer

A commercial explainer on reinforcement learning from AI feedback (RLAIF). It is useful for a nontechnical overview, but claims about bias propagation or performance should be replaced with primary RLAIF studies.

https://www.mercor.com/resources/experts/what-is-rlaif/

38

Awesome RLHF

Tier C GitHub resource list

A curated index of RLHF papers, libraries, and tutorials. It helps discover literature but is not itself evidence and should not be cited for substantive claims.

https://github.com/opendilab/awesome-RLHF

39

Reinforcement Learning from Human Feedback

Tier C RLHF Book

A detailed technical synthesis of preference data, reward modeling, optimization, and evaluation. It is a strong practitioner reference for the report's pipeline description, though specific empirical claims should point to the underlying studies.

https://rlhfbook.com/book.pdf

40

Linear Distortion of Unclipped RLHF at Every KL Budget

Tier B OpenReview PDF

A technical paper on worst-case distortion under RLHF-style aggregation. It is central to the report's warning that aggressive reward optimization can diverge from population welfare; interpret conclusions within the paper's formal assumptions.

https://openreview.net/pdf?id=PlspL3L46i

Preference optimization and RLHF (continued)

41

Distortion of AI Alignment Revisited: RLHF is a Decent Utilitarian

Tier A ICML 2026 poster page

The conference record corresponding to reference 16. It indicates a formal venue for the distortion analysis; the bibliography should merge the duplicate records and cite the final version.

https://icml.cc/virtual/2026/poster/60739

42

Distortion of AI Alignment: Does Preference Optimization Optimize for Average Reward?

Tier B arXiv:2505.23749 PDF

A primary technical preprint on whether preference optimization recovers average utility. It bears directly on the report's 'decent utilitarian' framing and should be compared carefully with references 16, 40, and 41 rather than summarized as a single settled result.

https://arxiv.org/pdf/2505.23749

43

Human-AI Collaboration with Misaligned Preferences

Tier B arXiv:2511.02746 PDF

A recent preprint on collaboration when human and AI objectives differ. It is relevant to preference mismatch and assistance, but the report should identify the exact model and avoid using the title alone as proof of real-world behavior.

https://arxiv.org/pdf/2511.02746

44

Individual Preferences and Group Objectives

Tier A UC Berkeley eScholarship paper

A scholarly treatment of moving from individual preferences to group objectives. It supports the report's social-choice framing and should help distinguish interpersonal comparison, aggregation rules, and fairness constraints.

https://escholarship.org/content/qt9rn2w1gd/qt9rn2w1gd.pdf

Preference optimization and RLHF (continued)

45

Aligning with Whom? A Neutrosophic Paraconsistent Framework

Tier B Preprints.org manuscript

An unreviewed 2026 manuscript proposing a formal response to conflicting values. It is potentially relevant to pluralistic alignment, but its novel framework should be presented as exploratory, not established practice.

https://www.preprints.org/manuscript/202607.2190

CIRL, decision theory, and changing rewards

46

AI Researcher Stuart Russell

Tier D Future of Life Institute profile

A biographical profile that can orient readers to Russell's role in AI safety. It does not substantiate the mechanics or guarantees of Cooperative Inverse Reinforcement Learning (CIRL).

https://futureoflife.org/ai-researcher-profile/ai-researcher-stuart-russell/

47

Cooperative Inverse Reinforcement Learning

Tier A arXiv:1606.03137

The foundational CIRL paper by Hadfield-Menell and colleagues. It formalizes assistance as a cooperative game with uncertainty about human reward and is the principal source for the report's claims about uncertainty and corrigibility.

https://arxiv.org/abs/1606.03137

48

Learning Human Utility from Video Demonstrations for Deductive Planning

Tier B CoRL-era paper PDF

An applied study of inferring human utility from demonstrations. It illustrates the practical promise and ambiguity of inverse learning; the report should not imply that observed behavior reveals fully informed preferences.

https://shukla.io/docs/2017_CoRL.pdf

49

Summary of Stuart Russell's Human Compatible

Tier D Effective Altruism Forum post

A secondary book summary useful for orientation to assistance-game ideas. Cite Russell's book or the CIRL paper for core claims, especially the off-switch argument.

https://forum.effectivealtruism.org/posts/tsHfFdAGehzoH6BZR/summary-of-stuart-russell-s-new-book-human-compatible

CIRL, decision theory, and changing rewards (continued)

50

Toward a Theory of Interactions Between Boundedly-Rational Agents

Tier B OpenReview PDF

A technical paper on agents with bounded rationality. It adds needed realism to idealized VNM-rational and perfectly revealing human models, but the report should specify which formal result it uses.

https://openreview.net/pdf?id=4Ft7DcrjdO

51

Pascal’s Mugging

Tier D LessWrong wiki

A concise community explanation of tiny-probability, enormous-utility pathologies. It is useful for introducing the thought experiment but should be paired with a primary decision-theory source for formal claims or proposed remedies.

https://www.lesswrong.com/w/pascal-s-mugging

52

Pascal’s mugging by Nick Bostrom [RT]

Tier D Reddit discussion

A discussion thread with no clear evidentiary contribution. It should be removed from an academic version unless it links to a recoverable primary text that is cited directly.

https://www.reddit.com/r/rational/comments/1wk90k/pascals_mugging_by_nick_bostrom_rt/

53

Notes on Pascal’s Mugging

Tier C Gwern.net essay

A long-form analytical essay collecting arguments and variants of the problem. It is more substantial than a forum thread, but remains secondary and should not be the sole basis for design recommendations.

https://gwern.net/mugging

CIRL, decision theory, and changing rewards (continued)

54

AI Alignment with Changing and Influenceable Reward Functions

Tier A PMLR-hosted paper PDF

A primary research paper on dynamic and influenceable rewards. It directly supports the report’s warning that an optimizing agent may have incentives to shape human preferences; this is one of the strongest technical sources in the final section.

https://raw.githubusercontent.com/mlresearch/v235/main/assets/carroll24a/carroll24a.pdf

Societal alignment, welfare, and fairness

55

Iason Gabriel

Tier D ResearchGate profile

An author profile, not a research source. It may help identify publications but should be removed from the bibliography once the relevant works are cited directly.

https://www.researchgate.net/profile/Iason-Gabriel

56

Dr Iason Gabriel

Tier D Personal website

An author website useful for publication discovery and biographical context. It does not independently support the report's normative taxonomy.

https://www.iasongabriel.com/

57

Societal Alignment Frameworks Can Improve LLM Alignment

Tier B OpenReview PDF

A technical submission testing societal-welfare frameworks for language-model alignment. It is highly relevant to the report's proposed combination of aggregate utility and protections for vulnerable groups; verify publication status and experimental scope.

https://openreview.net/pdf?id=HAXNSU8dZv

58

Democratizing value alignment: from authoritarian to democratic AI ethics

Tier B ResearchGate record

A paper on participatory or democratic value alignment. It supports the report's turn toward collective input, but the bibliography should cite the publisher or archival version and distinguish procedural legitimacy from technical performance.

https://www.researchgate.net/publication/386345791_Democratizing_value_alignment_from_authoritarian_to_democratic_AI_ethics

Societal alignment, welfare, and fairness (continued)

59

ACM SIGecom Exchanges, volume 24, issue 1

Tier C Issue PDF

A full journal issue rather than a specific article. The report should name the article, authors, and pages actually used; issue-level citation makes the supporting claim hard to audit.

https://www.sigecom.org/exchanges/volume_24/1/issue.pdf

60

Measuring inequality of well-being with a correlation-sensitive approach

Tier B ECINEQ working paper

A welfare-economics working paper relevant to inequality-sensitive aggregation. It can strengthen the move beyond simple sums of utility, but the report should explain how its welfare measure maps to AI objectives.

http://www.ecineq.org/milano/WP/ECINEQ2009-124.pdf

61

Atkinson-Stiglitz theorem in directed technological change

Tier D ResearchGate record

A highly specialized economic source whose connection to the report is unclear from the citation alone. Remove it unless the text explicitly uses a theorem or model from the paper and explains the relevance.

https://www.researchgate.net/publication/415066032_ATKINSON_STIGLITZ_THEOREM_IN_DIRECTED_TECHNOLOGICAL_CHANGE_DTC_AABH_MODEL_WITH_RAMSEY_MIRRLEES_PIGOU_POLICY_SYSTEM

62

Policy Design in Long-Run Welfare Dynamics

Tier B arXiv:2503.00632

A recent preprint on welfare over time. It is relevant to the report’s claim that short-run utility and long-run stability can diverge; any comparison between Rawlsian and utilitarian policies should reproduce the model assumptions and outcomes precisely.

https://arxiv.org/html/2503.00632

Societal alignment, welfare, and fairness (continued)

63

A guide to formulating fairness in an optimization model

Tier A PubMed Central

A practical scholarly guide to translating fairness concepts into optimization objectives and constraints. It is a strong methodological source for the report’s recommendation to combine welfare maximization with floors, constraints, or distribution-sensitive criteria.

https://pmc.ncbi.nlm.nih.gov/articles/PMC10081824/

The research spine supports a hybrid, constraint-aware view

The report's central synthesis is defensible as a research hypothesis: utility-based optimization is deeply embedded in modern AI, but welfare aggregation alone is insufficient for alignment. The bibliography becomes substantially stronger if it makes four revisions.

1. **Replace broad claims of inevitability with precise formal statements.** VNM representation, RL reward maximization, and moral utilitarianism are related but not identical. Cite the theorem, assumptions, and scope each time.
2. **Separate benchmark competence from deployed behavior.** ETHICS measures judgments over text; MACHIAVELLI studies goal pursuit in interactive fiction. Neither alone establishes safe or unsafe behavior in open-world systems.
3. **Treat preference aggregation as a social-choice design problem.** Use the distortion papers, group-objective work, and fairness optimization literature to specify whose utility is counted, how it is scaled, and what minority protections bind.
4. **Build the hybrid recommendation from primary evidence.** Ground uncertainty in CIRL, influence incentives in dynamic-reward research, and minimum-harm constraints in fairness and welfare modeling. Cite constitutional and democratic alignment work directly in a future revision.

High-value additions for a revised report

The current list would benefit from canonical primary sources on RLHF and InstructGPT, Direct Preference Optimization, Constitutional AI, social-choice impossibility and welfare theorems, Rawls's *A Theory of Justice*, and primary editions of Bentham and Mill. These additions would reduce dependence on explainers and make the final "hybrid alignment" recommendation traceable from first principles to modern implementations.

Bottom line. Use this bibliography as a map: the Tier A and strong Tier B items form the research spine; Tier C items explain or implement; Tier D items are discovery aids to replace, not evidence to lean on.